

SNO	Particulars	Theory		Practical	
		Amplitude	frequency	Amplitude	Time Frequency
1.	Message	5	400	5	3958 4045 = 1
2.	Carrier	12	1000	12	4045 = 1 4032.3
3.	Phase Modulation Signal	-	-	12	4045 = 1 4032.3 fmin = 4000 Hz fmax = 10000 Hz

Exp: 5
28/07/2026
Date:

SIMULATION OF PHASE MODULATION (PM) SIGNALS

Aim:

To generate and Simulate a Phase Modulated (PM) signal using Scilab and observe the message signal.

Apparatus Required:

- * Computer / Laptop
- * Scilab Software

CODE:

```
clc;  
clear;  
close;  
am = 5;  
fm = 400;  
ac = 12;  
fc = 4000;  
fs = 40000;  
t = 0:1/fs:2/fm;  
kp = 3.5  
em = am * cos(2 * %pi * fm * t);  
ec = ac * cos(2 * %pi * fc * t);
```

Calculation.

1. Message Signal

$$T = x_2 - x_1$$

$$= 3.74 \times 10^{-3} - 1.25 \times 10^{-3}$$

$$= 2.49 \times 10^{-3}$$

$$f_m = \frac{1}{T}$$

$$= \frac{1}{2.49 \times 10^{-3}}$$

$$= 401.6 \text{ Hz}$$

$$\Delta f = A_m k_p f_m$$

$$f_{\max} = f_c + 4 k_p f_m$$

$$f_{\min} = f_c - \Delta f$$

$$k_p = 3.5$$

$$\Delta f = 5 \times 3.5 \times 400$$

$$= 7000$$

$$f_{\min} = 3000 - 7000$$

$$= -4000$$

$$f_{\min} = 4000 \text{ Hz}$$

$$f_{\max} = 3000 + 4(3.5)400$$

$$= 8600 \text{ Hz}$$

2. Carrier Signal

$$T = x_2 - x_1$$

$$= (1.249 \times 10^{-3}) - (1.000 \times 10^{-3})$$

$$= 2.49 \times 10^{-4} \text{ s}$$

$$f_s = \frac{1}{T}$$

$$= \frac{1}{2.49 \times 10^{-4}}$$

$$f_{\max} = 4016.1 \text{ Hz}$$

3. Phase Modulation Signal

$$T = x_2 - x_1$$

$$= 3.748 \times 10^{-3} - 3.50 \times 10^{-3}$$

$$= 2.48 \times 10^{-4} \text{ s}$$

$$f_p = \frac{1}{T}$$

$$= \frac{1}{2.48 \times 10^{-4}}$$

$$f_{\min} = 4032.3 \text{ Hz}$$

$$e_{pm} = a_L \cos(2\pi \times 10^3 \pi f_c t + k_p m);$$

figure (1)

subplot(3,1,1);

plot(t, em);

xgrid();

title("Message Signal");

xlabel("Time(s)");

ylabel("Amplitude");

subplot(3,1,2)

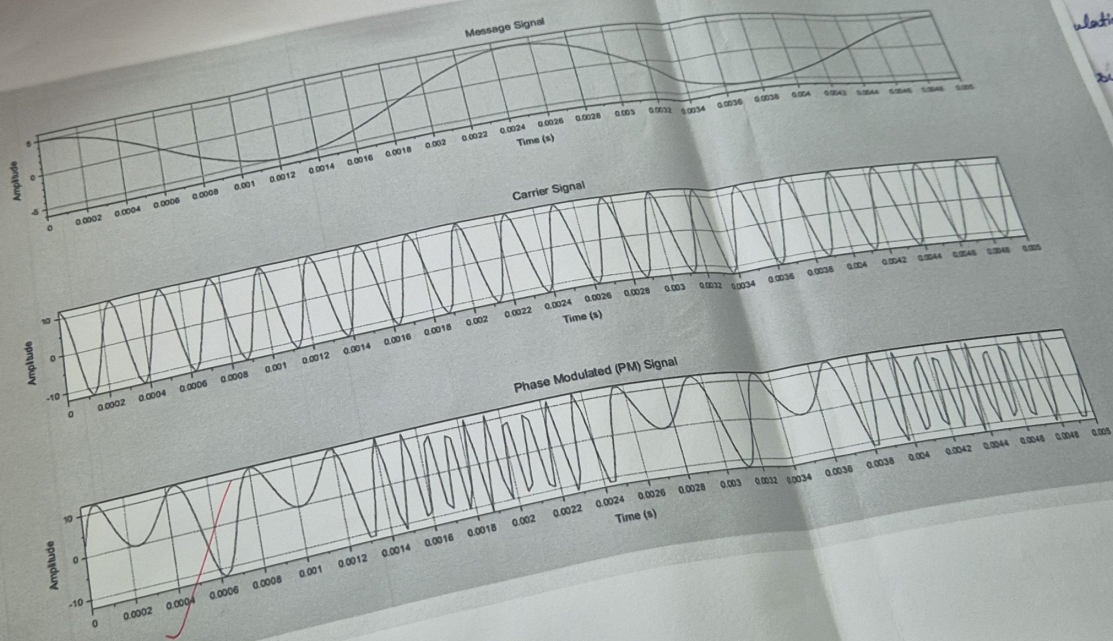
plot(t, ec);

xgrid();

title("Phase Modulator (PM) Signal");

xlabel("Time(s)");

ylabel("Amplitude");



	Mark	Mark Obtained
Generation	20	20
Modulation	20	20
Result	20	20
	20	16
	20	20
	20	20
Total		

Result:

The Phase Modulated signal is generated and simulated.

Sno	Content	Mark	Mark obtained
1.	Program implementation	20	20
2.	Waveform and tabulation	20	20
3.	Calculation and result	20	20
4.	Viva - Voice	20	16
5.	Result / observation	20	20
		Total	96

$$\frac{96}{5} = 19.2$$

$(M-x)$	$M-x$	
900.8	$900.8 - 2000.0 = -1099.2$	A
1100.1	$1100.1 - 2000.0 = -899.9$	F
1200.1	$1200.1 - 2000.0 = -800.0$	B
800.01	$800.01 - 2000.0 = -1199.99$	D
2000.0	$2000.0 - 2000.0 = 0.0$	C
2100.0	$2100.0 - 2000.0 = 100.0$	E

Result:

The Phase Modulated (PM) signal was successfully generated and simulated using Simulink

PSY